Nathan H. Carter

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EDUCATION

B.S. in Mathematics, Minors in Data Analytics & Computational Analytics 2024–2027
James Madison University, Harrisonburg, VA
Overall GPA: 3.91 | Math GPA: 3.92

A.A.S. in General Studies 2022–2024
Mountain Empire Community College, Big Stone Gap, VA
GPA: 4.00 (*Summa Cum Laude*, Earned concurrently via Dual Enrollment)

INDUSTRY & ENGINEERING EXPERIENCE

DevOps Engineer Intern 2025–2026
CGI, Lebanon, VA

– Managed lifecycle stability for four major repository forks within the DOTA (Tools and Automation) department; utilized Jenkins and Git to maintain high-integrity CI/CD pipelines for large-scale development branches.

Software Engineer Intern 2025
CGI, Lebanon, VA

– Engineered a Python-based automation framework for code analytics, achieving a 300x increase in processing speed and streamlining cross-departmental data reporting.

RESEARCH EXPERIENCE

Optimization & Algorithmic Theory Research Assistant Incoming Jan. 2027
Optimization & Algorithmic Theory Laboratory, James Madison University
Advisor: Dr. Zev Woodstock

– Incoming position to continue concurrent proximal relaxation averaging project and possible new research within the laboratory.

Mathematics Researcher 2026
Optimization & Algorithmic Theory Laboratory, James Madison University
Advisor: Dr. Zev Woodstock

– Investigated the stability of proximal average relaxation algorithms for signal recovery under measurement noise and linear/nonlinear operator perturbations.

– Formulated explicit quantitative sensitivity bounds demonstrating linear error scaling and validated theoretical results using Tikhonov-regularized image reconstruction models.

Undergraduate Mathematics Researcher 2026
Department of Mathematics and Statistics, James Madison University
Advisor: Dr. Hasan Hamdan

– Compared Bayesian Gibbs Sampling, Expectation-Maximization (`mclust`), and UNMIX scale mixtures for modeling daily log-returns of top U.S. equities.

– Conducted in-sample and out-of-sample density forecasts to analyze model trade-offs between empirical precision and generalization across market regimes.

Undergraduate Educational Researcher 2025
College of Education, James Madison University
Advisor: Dr. Angela Webb

– Developed a strong foundation in qualitative research design, ethics, and academic writing to complement quantitative modeling methodologies.

AWARDS AND HONORS

- **Centennial Scholars Program**, Merit-based full-tuition scholarship 2024–2027
- **Klippert & Mills Math Scholarship**, James Madison University 2026
- **Jeffrey E. Tickle Scholarship in Mathematics**, James Madison University 2026
- **Mathematics Department Scholarship**, James Madison University 2026
- **Phi Kappa Phi Honor Society** (Top 7.5% of Junior Class) 2026
- **President's List**, James Madison University (GPA $\geq$ 3.90) 2024–2026
- **FYRE Research Fellowship**, First-Year Research Experience 2025
- **Summa Cum Laude**, Mountain Empire Community College (4.0 GPA) 2024

PUBLICATIONS & MANUSCRIPTS IN PREPARATION

- **N. Carter**, K. Scafe, & Z. Woodstock. *Stability of Proximal Average Relaxation for Signal Recovery*. (Manuscript in preparation).
- **N. Carter** & H. Hamdan. *Using Scale Mixtures of Normals to Model Continuously Compounded Returns*. (Manuscript in preparation).

TEACHING & TUTORING EXPERIENCE

Learning Assistant (MATH 205: Applied Calculus) 2026
Department of Mathematics and Statistics, James Madison University

– Assisted during active-learning class sessions, guiding students through problem sets.
– Held weekly office hours to provide one-on-one instruction.

Mathematics Tutor 2024–2025
World Tutors

– Provided remote mathematics instruction and homework support for students in grades 1–9.

COURSEWORK & TECHNICAL SKILLS

- **Mathematics Coursework:** Advanced Calculus I (Analysis), Abstract Algebra I, Advanced Linear Algebra, Modern Optimization, Differential Equations, Topology, Graph Theory, Discrete Math, *Advanced Calculus II (Upcoming)*, *Operations Research (Upcoming)*
- **Statistics Coursework:** Applied Linear Regression, Machine Learning, Data Analysis & Visualization, Mixture Modeling, Introduction to Data Science
- **Programming Languages:** Python, R, Java, Bash, Julia
- **Tools & Software:** LaTeX, Beamer, Git, GitHub, Linux/Unix, Vim, Jenkins CI/CD

PROFESSIONAL AFFILIATIONS

- **Mathematical Association of America (MAA)** 2026–Present
- **American Statistical Association (ASA)** 2026–Present
- **The Honor Society of Phi Kappa Phi** 2026–Present

PRESENTATIONS & POSTERS

Shenandoah Undergraduate Mathematics and Statistics Conference November 2026
Stability of Proximal Average Relaxation for Signal Recovery (Undergraduate Talk), Harrisonburg, VA

James Madison University CSM Summer Symposium July 2026
Stability of Proximal Average Relaxation for Signal Recovery (Undergraduate Talk), Harrisonburg, VA

James Madison University Student Colloquium May 2026
Normal Mixture Density Estimation for Major US Stocks: A Four-Year Empirical Study (Undergraduate Talk), Harrisonburg, VA

M.A.A. MD-DC-VA Section Meeting April 2026
Normal Mixture Density Estimation for Major US Stocks: A Four-Year Empirical Study (Contributed Talk), Frederick, MD

James Madison University FYRE Symposium October 2025
A Comprehensive Examination of University-Based Induction Support for Early Career Teachers (Poster Presentation), Harrisonburg, VA